Chapter: 3

Chemical Bonding

1) Which of the following occurs in an ionic bond?
a. Oppositely charged ions attract.
b. Two atoms share two electrons.
c. Two atoms share more than two electrons.
d. Like-charged ions attract.

2) How does Magnesium (Z=12) get octet configuration when reacting to form compounds?
a. It gains electrons.
b. It gives up electrons.
c. It does not change its number of electrons.
d. Calcium does not obey the octet rule.

3) How many valence electrons are transferred from the Aluminium atom to iodine atoms in the formation of the compound Aluminium iodide?
a. 0 b. 1 c. 2 d. 3

4) Which of the following pairs of elements is most likely to form an ionic compound?
a. magnesium and fluorine
b. nitrogen and sulfur
c. oxygen and chlorine
d. sodium and aluminum

5) From the following , which is most stable electron configuration?
(2,3 2,8,2, 2,8 2,8,1)

6) Electronegativity of Fluorine on the Pauling Electronegativity Scale
(3, 4, 5, 6)

7) Highest value on the Pauling electronegativity scale?
(3, 4, 5, 6)

8) The element with the lowest electronegativity.
(Fluorine, Francium, Hydrogen, Oxygen)

9) Element with highest electronegativity.
(Fluorine, Francium, Hydrogen, Oxygen)

10) When atoms combine to form molecules, ---------- transfers or shares between the atoms.
(Proton, Neutron, Electron)

11) When an atom gains an electron and becomes a negative ion.
(Energy received, energy released, no change in energy.)

12) $Na \rightarrow Na^+ + 1e^-$
According to this sodium atom ..... energy.
(Accepts, emits, does not change.)

13)An atom accepts an electron and releases energy when it becomes a negative ion. This energy is called .....
(Ionization energy, electron gain enthalpy, ionization enthalpy)

14) charge of cations is ----------
(Positive, negative, no charge)

15) The electrostatic force of attraction that holds together the oppositely charged ions in an ionic compound is called ----------

16) The electron configuration of X is 2, 8, 7 electrons. It combines with Y having 1 electron in its outermost shell.
(a) What type of bond will be formed between X and Y?
(b) Write the formula of the compound formed.

17) The electron configuration of fluoride ion is the same as that of a neon atom. Write one similarity and one difference between the two?

18) The element X has the electronic configuration 2, 8, 18, 8, 1. Without identifying x,
(a) predict the sign and charge on a simple ion of X.

19) How do atoms attain noble gas configuration?

20) In the formation of the compound XY2, an atom X gives one electron to each Y atom. What is the nature of bond in XY2? Draw the electron dot structure of this compound.

21) Is an electron configuration 2,8,7 is stable? Why?

22) Draw orbit structure and electron dot diagram of NaCl, $MgCl_2$ and CaO.

23) What is Electronegativity?

24) Add the symbols δ^- and δ^+ to the polar molecules given below.
HCl, NH_3, H_2O

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25) Complete the activity given below and then write the chemical formula of Aluminiun fluride. Write the constituent elements of aluminum fluoride.
Write the atomic number and electron configuration of aluminum and fluorine.
Find the valency of aluminum and fluorine.
Write the symbols in the order that the element with the least electronegativity comes first.
Write the chemical formula.

26) The chemical formula magnesium chloride is $MgCl_2$ and sodium oxide is Na_2O. Find the valency Mg, Cl, Na, and O.
Then write Chemical Formula
a) Magnesium oxide.
b) sodium chloride.

27) What is neutralization reaction?

28) Correct the mistake in the given sentence.
a) Hydrochloric acid [HCl] is a dibasic acid.
b) The basicity of phosphoric acid [H_3PO_4] is 2.
c) Carbonic acid is a dibasic acid. So the chemical formula is H_3CO_3.

29) When hydrochloric acid and sodium hydroxide react together
a) Which salt is obtained?
b) Write the chemical equation of it.

30) When phosphoric acid (H_3PO_4) reacts with magnesium hydroxide [$Mg(OH)_2$]
a) Write the chemical formula of the resulting salt.
b) Write the chemical equation of of reaction.

31) Complete the table with the help of the periodic table.

Compound	Constituent elements	Atomic number	Electron configuration	The number of electrons shared or transferred by each atom	Valency
NaCl	Na	11	2, 8, 1	1	1
	Cl	17	2,8,7	1	1
MgO	Mg				
	O				

32) Illustrate and write the chemical equation for the sodium atom to become an ion.

33) Write the electron configuration of chlorine and find how many electrons are needed to complete the octate electron configuration of chlorine.

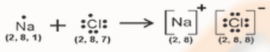
34) Prepare the chemical equation and illustration of how a chlorine atom accepts an electron to become chloride.

35) What is electron gain enthalpy?

36) 

Electron dot diagram of chlorine is given. Draw the electron dot diagram of sodium.

37) The electron dot diagram of sodium chloride formation is given. Use it and write the answers to the questions given below.

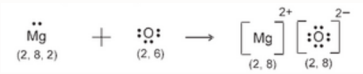


Give detals of sodium atom before chemical reaction.
i) Number of electrons.
ii) Number of protons
iii charge of atom

38) Classify the symbols given in the box into cations, anions and atoms.

Na^+, Cl^-, Ca, Mg^{2+}, O^{2-}, He

39) Examine the given electron dot diagram and write the answers to the questions.



a) Which is the anion in the formation of magnesium oxide?
b) Which element donates electrons in the formation of magnesium oxide?
c) What is the number of electrons shared by magnesium to form magnesium oxide?

40) Choose the appropriate statements for ionic compounds from the given ones.
i) They are generally insoluble in water.
ii) They are non volatile and hard.
iii) Covalent compounds exist in solid, liquid and gaseous states.
iv) They are not conductors of electricity in solid state, they conduct electricity in molten and aqueous states They are generally insoluble in water.

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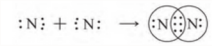
41) The electron dot diagram of the fluorine molecule is given. Related to this
Write the answers to the questions.
a) How many more electrons does fluorine need to have an octet configuration?
b) How many electrons are shared by a fluorine atom?
c) How many pairs of electrons are shared in a chemical bond in a fluorine molecule?
d) Indicate by symbols the single bond between atoms in fluorine molecule.

42) The electron dot diagram of the oxygen molecule is given. Related to this
Write the answers to the questions.



a) How many more electrons does oxygen need to have an octet configuration?
b) How many electrons are shared by a oxygen atom?
c) How many pairs of electrons are shared in a chemical bond in a oxygen molecule?
d) Indicate by symbols the double bond between atoms in oxygen molecule.

43) The electron dot diagram of the nitrogen molecule is given. Related to this
Write the answers to the questions.



a) How many more electrons does nitrogen need to have an octet configuration?
b) How many electrons are shared by a nitrogen atom?
c) How many pairs of electrons are shared in a chemical bond in a nitrogen molecule?
d) Indicate by symbols the triple bond between atoms in nitrogen molecule.

44) What is the positive ion formed when an acid dissolves in water?

45) What is the negative ion/radical formed when alkali dissolves in water?

46) The negative ion in hydrochloric acid is 'Cl-' and basicity is 1 then the chemical formula of then write chemical formula of the acid.

47) The negative ion in sulphuric acid is 'SO42-' and basicity is 2 then write chemical formula of the acid.

48) The negative ion in phosphoric acid is 'PO43- ' and basicity is 3 then write chemical formula of the acid.

49) The negative ion in Nitric acid is 'NO3-' and basicity is 1 then write chemical formula of the acid.

50) The positive ion of the alkali sodium hydroxide is Na+. Write the chemical formula.

51) The positive ion of the alkali sodium hydroxide is K+. Write the chemical formula.

52) The positive ion of the alkali calcium hydroxide is Ca2+. Write the chemical formula.

53) The positive ion of the alkali aluminum hydroxide is Al3+. Write the chemical formula.

54) The positive ion of the alkali ferric hydroxide is Fe3+. Write the chemical formula.

55) What are the constituent element of sodium chloride?

56) Write the electron configuration of sodium [Na-11] and write the way to obtain octave electron configuration.

57) What is the number of electrons given up by the sodium atom while participating in the chemical reaction?

58) Illustrate and write the chemical equation for the sodium atom to become an ion.

59) What are cations?

60) What are anions?

61) What is ionic bonding?

62) What is the product when magnesium ribbon burns in air?

63) Complete the table with the help of the periodic table.

HF	H				
	F				
CCl_4	C	6	2, 4		
	Cl	17	2, 8, 7		
BF_2	B				
	F				
H_2O	H				
	O				

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64) Poling electronegativity scale values of certain elements are given.
(Na-0.93 Ca-1 Mg-1.31 O-3.44 Cl-3.16 H- 2.20 F- 4)
Complete the table.

Compounds	Difference in electronegativity of constituent elements	Nature of compounds (ionic / covalent)
Sodium chloride - NaCl	3.16-0.93 = 2.23	ionic
Hydrogen chloride-HCl		
Sodium oxide-Na_2O		
Methane- CH_4		
Magnesium fluoride-MgF_2		

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